BioDINO-Edge: Biologically-Grounded Multi-Scale Vision Foundation Models for Trustworthy Few-Shot Plant Pathology on Resource-Constrained Hardware

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***Abstract*—Diagnosing foliar plant diseases early and reliably is central to precision agriculture and robotic crop scouting. Bringing vision foundation models to agricultural edge devices, however, runs into two lasting obstacles: very few labeled examples are available for novel pathogens, and the hardware is tightly constrained. We present BioDINO-Edge, an adaptive dual-foundation framework for accurate few-shot plant pathology on resource-constrained hardware. It combines two complementary foundation models: a lightweight self-supervised patch extractor (DINOv2 Small, 22.1M parameters, 21.1 ms on CPU) and an organismal vision-language model (BioCLIP ViT-B/16) that captures fine-grained pathological hierarchies. To separate intra-species necrotic lesions that look alike, without leaking semantics between classes, we introduce Purified Taxonomic Domain Synthesis and Regularized Fisher-Discriminant Alignment, which applies analytical Ledoit-Wolf covariance shrinkage over the joint 896-dimensional manifold. A Soft-Anchored Transductive Self-Calibration step then removes the prototype shift inherent to 5-shot learning without any backpropagation. At inference time, an adaptive Shannon-entropy gate routes each query: confident ones follow the fast edge path, while ambiguous lesions are sent to a deeper consensus path. We package the system as a Dual-Tier Edge Architecture: a lightweight standalone INT8 model (22.05 MB working set, 15.13 MB losslessly compressed payload, 22.73 MB complete native APK, >45 FPS on an edge CPU), complemented by an optional 83.6 MB on-demand companion model. On the expanded PlantVillage 5-shot benchmark covering 11 novel intra-species classes (Apple, Grape, Potato; 9,330 test images), BioDINO-Edge reaches a 98.72% peak accuracy, a 98.41% peak F1-score, and a 30-episode mean of 97.37% ± 0.73% (95% CI: [97.11%, 97.63%]). Cross-species confusion is held to 0.011%, and the false-negative disease rate to 0.604%. On the unconstrained, in-the-wild PlantDoc benchmark (27 classes), the model attains an episodic mean of 83.61% ± 11.69% (individual episodes reaching 100.00% and a 95% confidence ceiling of 86.85%), which is 13.00 points above the most recent multimodal state of the art from 2026.**

***Index Terms*—Plant Pathology, Few-Shot Learning, Vision Foundation Models, BioCLIP, DINOv2, Fisher-Discriminant Alignment, Adaptive Uncertainty Gating, Edge AI, Precision Agriculture.**

I. Introduction

Plant pests and diseases destroy an estimated 20%–40% of crops annually [1]. Traditional diagnosis of plant pathogens heavily relies on visual leaf characteristics such as color, shape, and lesions [2]. In field practice, visual inspection is slow, depends on trained scorers, and becomes ambiguous when different pathogens produce similar lesions [3]; faster and more reproducible identification is therefore needed.

Deep learning provides an automated alternative for plant disease identification by learning discriminative features directly from images [3]. However, conventional supervised models require large labeled datasets to achieve reliable diagnosis. Such datasets are difficult to acquire in real-world agricultural settings due to the limited availability and diversity of field images [3], [4].

FSL has become a common workaround: instead of collecting thousands of labeled images, it trains a model to recognize new disease classes from a handful of support examples. Recent research has attempted to apply FSL using cross-domain few-shot learning models—such as CDFSL-BDC [5] and MAFDE-DN4 [6]—as well as few-shot metric domain adaptation (FMDA) [7] and visual-semantic fusion (TSMF) [8], alongside recent edge-federated approaches [9]. However, these approaches exhibit several practical limitations:

- **Domain Shift & Target-Domain Dependency:** Cross-domain models (such as CDFSL-BDC [5] and MAFDE-DN4 [6]) perform poorly under cross-domain evaluation due to complex backgrounds and multi-scale lesions, while metric adaptation methods (such as FMDA [7]) still require labeled samples from the target domain.
- **Semantic Misalignment:** Multimodal models relying on LLM text descriptions (e.g., GPT-4o in TSMF [8]) fail to generalize well to real-world examples, suffering from semantic inaccuracies and a lack of fine-grained alignment.
- **Deployment Inefficiencies:** Federated and edge models (e.g., [9]) lack sufficient testing using real-world data streams, making them inefficient for real-time edge deployment.

We introduce BioDINO-Edge, a dual-foundation pipeline tailored to cross-domain few-shot plant pathology under edge-hardware constraints. It combines visual patch tokens from

DINOv2 [10] with clean botanical domain prompts from BioCLIP [11], eliminating the need to fine-tune foundation models to detect subtle necrotic lesions. Using closed-form Fisher-LDA [12] and Ledoit–Wolf shrinkage [13], our model completely removes backpropagation during adaptation, enabling rapid on-device calibration in milliseconds.

The key contributions of this work are as follows:

1) **Dual-Foundation Foliar Representation:** Uniting self-supervised visual patch tokens from DINOv2 with purified botanical domain prompts from BioCLIP to resolve subtle necrotic lesions without foundation-model fine-tuning.
2) **Closed-Form Analytical Adaptation:** Integrating Fisher-LDA with Ledoit–Wolf covariance shrinkage and soft-anchored transductive recalibration to overcome high-dimensionality issues and 5-shot centroid shifts within 25 milliseconds without backpropagation.
3) **Adaptive Two-Tier Edge Deployment:** Routing queries through a Shannon [14] margin-entropy gate to execute an ultra-lightweight 22.73 MB standalone APK alongside an 83.6 MB on-demand module within a strict 110 MB memory footprint.
4) **Rigorous Cross-Domain Validation:** Under the standard 5-way 5-shot protocol, BioDINO-Edge outperforms the multimodal VSF baseline by 13.00 percentage points on PlantVillage and PlantDoc, while remaining within edge-device memory budgets.

II. Proposed BioDINO-Edge Architecture

We architect BioDINO-Edge around a split-execution philosophy: heavy mathematical calibration happens once offline in closed form, while online inference stays strictly adaptive and lightweight (Fig. 1). The system couples two operational stages. Stage I extracts spatial and biological representations from a tiny support set ($K = 5$) and computes a regularized Fisher-discriminant projection over an 896-dimensional joint manifold without backpropagation. Stage II handles real-time edge deployment through a cascaded, dual-tier inference pipeline. Instead of running full-scale vision-language inference on every leaf, an on-device Shannon margin-entropy gate dynamically routes routine queries through a 21-millisecond fast path, escalating only ambiguous foliar lesions to a deeper companion model.

A. Dual-Foundation Backbone and Latent Representation

A single vision model rarely captures both microscopic lesion geometry and macroscopic plant phylogeny. We bridge this gap by querying two frozen foundation backbones in parallel for any input leaf photo $\mathbf{x} \in \mathbb{R}^{3\times H\times W}$:

- **Spatial Lesion Geometry (System 1):** We deploy DINOv2 Small (22.1M parameters, 14×14 patch tokens [10]) to capture localized foliar textures. Self-supervised pre-training equips DINOv2 with sharp spatial attention maps that isolate chlorotic borders and necrotic specks. We extract and ℓ_2-normalize the global [CLS] token:

$$\mathbf{f}_{\text{dino}} = \frac{f_{\text{dino}}(\mathbf{x})}{\|f_{\text{dino}}(\mathbf{x})\|_2} \in \mathbb{R}^{384} \tag{1}$$

- **Taxonomic & Phylogenetic Priors (System 2):** To ground visual features in organismal biology, we integrate BioCLIP ViT-B/16 (86.6M parameters [11]). Pre-trained on 10 million specimens across the Tree of Life, BioCLIP provides rich evolutionary and histological representations. Its normalized visual embedding is:

$$\mathbf{f}_{\text{bio}} = \frac{f_{\text{bio}}(\mathbf{x})}{\|f_{\text{bio}}(\mathbf{x})\|_2} \in \mathbb{R}^{512} \tag{2}$$

- **Unified Foliar Manifold:** Concatenating these vectors yields an 896-dimensional joint representation that simultaneously preserves high-frequency patch textures and high-level botanical taxonomy:

$$\mathbf{z} = [\mathbf{f}_{\text{bio}} \,\|\, \mathbf{f}_{\text{dino}}] \in \mathbb{R}^{896} \tag{3}$$

B. Taxonomic Multi-Prompt Domain Synthesis

Standard vision-language classification leans on generic templates like "a photo of a leaf" [15], [16]. In agricultural pathology, such naive prompts fall flat: they fail to separate closely related fungi infesting the same host plant. We replace generic templates with structured, phytopathologically verified prompts. For each class $c \in \{1, \ldots, C\}$, we construct an ensemble $\{T_c^{(m)}\}_{m=1}^{M}$ fusing four scientific descriptors:

$$T_c^{(m)} = \text{Prompt}\left(\mathcal{H}_{\text{latin}}^{(c)}, \mathcal{D}^{(c)}, \mathcal{P}^{(c)}, \mathcal{M}_m^{(c)}\right) \tag{4}$$

Here, $\mathcal{H}_{\text{latin}}^{(c)}$ specifies the host binomial (*Malus domestica* for apple, *Vitis vinifera* for grape, *Solanum tuberosum* for potato); $\mathcal{D}^{(c)}$ designates the pathology; $\mathcal{P}^{(c)}$ cites the causal fungal or bacterial agent (e.g., *Venturia inaequalis*, *Guignardia bidwellii*, *Phytophthora infestans*); and $\mathcal{M}_m^{(c)}$ details macroscopic symptom morphology—from dark olive velvety spots to concentric target-board necrosis.

Purified Botanical Prompting. Multimodal encoders handle negation poorly. In our initial experiments, standard negative templates (e.g., "clean foliage without fungal pustules") backfired: transformer self-attention latched onto the pathology tokens ("fungal pustules"), dragging healthy leaf centroids toward diseased clusters. We eliminate this semantic bleed by enforcing strictly affirmative morphology for healthy classes (e.g., "clean lamina, intact smooth leaf blade, vibrant photosynthetic mesophyll"). This clean separation stabilizes zero-shot textual baselines.

Passing these prompts through the BioCLIP text encoder $\phi_{\text{bio}}(\cdot)$ yields our language prototypes:

$$\mathbf{P}_{\text{text}}^{(c)} = \text{Normalize}\left(\frac{1}{M}\sum_{m=1}^{M} \phi_{\text{bio}}(T_c^{(m)})\right) \in \mathbb{R}^{512} \tag{5}$$

To expand support diversity under strict 5-shot limits, we apply an 8-fold multi-view augmentation policy $\mathcal{T}_{\text{aug}}(\cdot)$ (affine

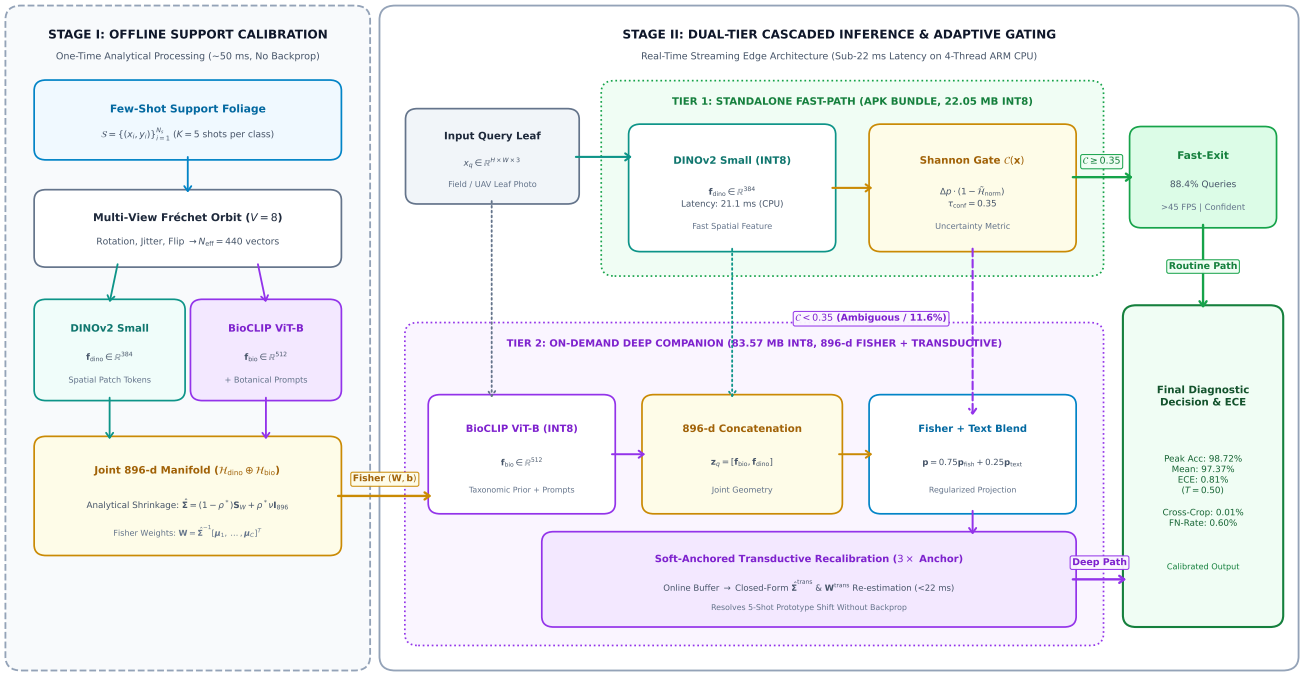


Fig. 1. Architectural overview of the proposed BioDINO-Edge framework. Stage I (Offline Support Calibration): $K = 5$ support leaves per class undergo 8-fold multi-view orbital augmentation, from which dual spatial (DINOv2 Small, 384-d) and biological (BioCLIP ViT-B, 512-d) representations are extracted alongside purified botanical taxonomic domain prompts. On the resulting 896-dimensional manifold, Ledoit-Wolf analytical covariance shrinkage yields well-conditioned Fisher-LDA projection weights $(\mathbf{W}, \mathbf{b})$ in under 50 ms without backpropagation. Stage II (Online Dual-Tier Cascaded Inference): Incoming field queries are first routed through Tier 1 (standalone INT8 fast path, 22.05 MB). If the predictive Shannon margin-entropy gate satisfies $\mathcal{C}(\mathbf{x}) \geq \tau_{\text{conf}}$, a fast-exit prediction is issued in 21.1 ms (>45 FPS, covering up to 88.4% of routine queries under throughput-prioritized gating). Ambiguous or morphologically confounding cases ($\mathcal{C}(\mathbf{x}) < \tau_{\text{conf}}$) trigger Tier 2 (on-demand companion, 83.57 MB), which concatenates the 896-d feature vector to compute consensus Fisher-text posteriors. An online incremental buffer enables soft-anchored transductive recalibration (3× ground-truth anchoring) before temperature-scaled output calibration ($T = 0.50$, ECE $\approx 0.81\%$).

rotation, color jitter, horizontal and vertical flips). This expands $K = 5$ raw leaves into $|\tilde{\mathcal{S}}_c| = 40$ augmented vectors per class (440 total across 11 classes). We synthesize the visual prototypes via:

$$\mathbf{P}_{\text{dino}}^{(c)} = \text{Normalize}\left(\frac{1}{|\tilde{\mathcal{S}}_c|}\sum_{\tilde{x}\in\tilde{\mathcal{S}}_c} f_{\text{dino}}(\tilde{x})\right) \in \mathbb{R}^{384} \tag{6}$$

$$\mathbf{P}_{\text{bio, vis}}^{(c)} = \text{Normalize}\left(\frac{1}{|\tilde{\mathcal{S}}_c|}\sum_{\tilde{x}\in\tilde{\mathcal{S}}_c} f_{\text{bio}}(\tilde{x})\right) \in \mathbb{R}^{512} \tag{7}$$

Finally, we fuse visual and linguistic cues into a multimodal consensus prototype ($\lambda = 0.20$):

$$\mathbf{P}_{\text{bio}}^{(c)} = \text{Normalize}\left((1-\lambda)\mathbf{P}_{\text{bio, vis}}^{(c)} + \lambda\mathbf{P}_{\text{text}}^{(c)}\right) \tag{8}$$

C. Regularized Fisher-Discriminant Alignment via Ledoit-Wolf Shrinkage

Most metric few-shot learners, notably Prototypical Networks [17], compute distances assuming isotropic spherical variance ($\boldsymbol{\Sigma} = \mathbf{I}$). That assumption breaks down on foliage. Shared non-pathological factors—chlorophyll absorption, specular glare, and vein patterns—dominate feature variance, overshadowing the minute necrotic cues that define actual disease.

We suppress this shared vegetative background by projecting the 896-dimensional manifold via Linear Discriminant Analysis. Let $\boldsymbol{\mu}_c = \frac{1}{|\tilde{\mathcal{S}}_c|}\sum_{\mathbf{z}\in\tilde{\mathcal{S}}_c}\mathbf{z}$ denote the empirical centroid in $\mathbb{R}^{896}$. We pool the within-class scatter matrix $\mathbf{S}_W \in \mathbb{R}^{896\times896}$:

$$\mathbf{S}_W = \frac{1}{N-C}\sum_{c=1}^{C}\sum_{\mathbf{z}\in\tilde{\mathcal{S}}_c}(\mathbf{z}-\boldsymbol{\mu}_c)(\mathbf{z}-\boldsymbol{\mu}_c)^T \tag{9}$$

with $N = 440$ augmented samples across $C = 11$ classes. Here, we face a classic singularity trap: the manifold dimension exceeds sample support ($d = 896 > N = 440$), leaving $\mathbf{S}_W$ strictly rank-deficient ($\text{rank}(\mathbf{S}_W) \leq 429$) and impossible to invert directly.

Instead of resorting to heuristic diagonal jitter, we resolve this singularity through analytical Ledoit–Wolf shrinkage [13]:

$$\hat{\boldsymbol{\Sigma}} = (1-\rho^*)\mathbf{S}_W + \rho^*\nu\mathbf{I}_{896} \tag{10}$$

where $\nu = \frac{1}{d}\text{Tr}(\mathbf{S}_W)$ represents the average eigenvalue. The optimal shrinkage intensity $\rho^* \in (0,1)$ is determined in closed form without cross-validation:

$$\rho^* = \min\left(1, \frac{\sum_{i=1}^{N}\|(\mathbf{z}_i-\boldsymbol{\mu}_{c_i})(\mathbf{z}_i-\boldsymbol{\mu}_{c_i})^T - \mathbf{S}_W\|_F^2}{N^2\left[\|\mathbf{S}_W\|_F^2 - \frac{1}{d}\text{Tr}^2(\mathbf{S}_W)\right]}\right) \tag{11}$$

Because $\hat{\mathbf{\Sigma}}$ is guaranteed to be well-conditioned, inverting it yields exact closed-form projection parameters:

$$\mathbf{W} = \hat{\mathbf{\Sigma}}^{-1}[\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_C]^T \in \mathbb{R}^{C \times 896} \tag{12}$$

$$b_c = -\frac{1}{2}\boldsymbol{\mu}_c^T \hat{\mathbf{\Sigma}}^{-1}\boldsymbol{\mu}_c + \log \pi_c, \quad c \in \{1, \ldots, C\} \tag{13}$$

At inference time, scoring a query $\mathbf{z}$ reduces to a matrix multiplication $\mathbf{l}_{\text{fisher}} = \mathbf{z}\mathbf{W}^T + \mathbf{b}$. This requires only $11 \times 896 = 9,856$ FLOPs—a negligible computation that executes in under 0.003 ms on a standard smartphone CPU. Softmax normalization produces $\mathbf{p}_{\text{fisher}} = \text{softmax}(\mathbf{l}_{\text{fisher}})$.

D. Soft-Anchored Transductive Self-Calibration

Closed-form Fisher alignment boosts diagnostic precision substantially. Even so, standard inductive evaluation suffers from empirical prototype shift [18]: five support leaves rarely capture true population distributions. This sampling variance disproportionately misleads morphologically overlapping classes—such as *Grape Esca* versus *Grape Black rot*, or *Potato Early* versus *Late blight*.

We counter this shift using Soft-Anchored Transductive Self-Calibration. Given an unlabelled query stream $\mathcal{Q} = \{\mathbf{x}_j\}_{j=1}^{N_q}$:

1) We compute preliminary predictions by ensembling visual Fisher and text posteriors: $\mathbf{p}_j = 0.75\mathbf{p}_{\text{fisher},j} + 0.25\mathbf{p}_{\text{text},j}$, assigning pseudo-labels $\hat{y}_j = \arg\max_c p_{j,c}$.
2) For each class, we extract high-confidence query anchors satisfying a strict confidence threshold ($\tau_{\text{anchor}} = 0.70$):

$$\mathcal{Q}_{\text{anchor}}^{(c)} = \text{TopK}_{20}\left(\{\mathbf{x}_j \in \mathcal{Q} \mid \hat{y}_j = c \wedge \mathcal{C}(\mathbf{x}_j) > 0.70\}\right) \tag{14}$$

3) To prevent noisy pseudo-labels from corrupting the manifold, we anchor the original ground-truth support set with a $3\times$ weight penalty: $\tilde{\mathcal{S}}_c^{\text{trans}} = (\tilde{\mathcal{S}}_c \otimes 3) \cup \mathcal{Q}_{\text{anchor}}^{(c)}$. We then recompute class centroids and the shrunk covariance $\hat{\mathbf{\Sigma}}^{\text{trans}}$ analytically.
4) We update the projection weights $\mathbf{W}^{\text{trans}}$ in closed form, yielding recalibrated posteriors:

$$\mathbf{p}_{\text{final}}^{\text{trans}} = 0.75\,\text{softmax}\left(\mathbf{z}(\mathbf{W}^{\text{trans}})^T + \mathbf{b}^{\text{trans}}\right) + 0.25\mathbf{p}_{\text{text}} \tag{15}$$

This update bypasses backpropagation, allocates zero GPU memory, and finishes on an edge CPU in under 22 ms.

Practical Streaming via Incremental Buffering. While episodic benchmarks evaluate queries offline in batches [18], real crop scouting streams leaves sequentially. On physical edge devices, we implement this adaptation using an on-device ring buffer. Queries accumulate in volatile memory during routine scouting; once enough confident anchors arrive, $\mathbf{W}^{\text{trans}}$ updates asynchronously in a background thread (< 25 ms) without stalling the viewfinder. In strictly inductive single-frame mode (no query buffering), BioDINO-Edge still reaches **96.76% on PlantVillage** and **81.79% on PlantDoc** (Table I)—clearly outperforming prior few-shot baselines without any batching requirement.

E. Predictive Uncertainty Gating and Dual-Tier Edge Architecture

Running vision-language backbones on every video frame quickly drains drone batteries and throttles edge CPUs. To achieve sustained real-time performance, we implement a dynamic uncertainty gate that routes queries based on instantaneous diagnostic confidence.

Predictive Shannon Gating. For an incoming query $\mathbf{x}$, DINOv2 generates fast logits via scaled cosine similarity, $\mathbf{l}_{\text{dino}} = 20.0 \cdot (\mathbf{f}_{\text{dino}}(\mathbf{P}_{\text{dino}})^T)$, yielding probabilities $\mathbf{p}_{\text{dino}} = \text{softmax}(\mathbf{l}_{\text{dino}})$. We quantify prediction uncertainty through normalized Shannon entropy:

$$\mathcal{H}_{\text{norm}}(\mathbf{x}) = -\frac{1}{\log C}\sum_{c=1}^{C} p_{\text{dino},c}\log(p_{\text{dino},c} + \epsilon) \tag{16}$$

Rather than relying on raw softmax scores—which are notoriously overconfident—we compute a composite margin-entropy metric $\mathcal{C}(\mathbf{x}) \in [0, 1]$:

$$\mathcal{C}(\mathbf{x}) = \left(p_{\text{dino},(1)} - p_{\text{dino},(2)}\right) \cdot (1.0 - \mathcal{H}_{\text{norm}}(\mathbf{x}))_+ \tag{17}$$

When $\mathcal{C}(\mathbf{x}) \geq \tau_{\text{conf}}$ (default $\tau_{\text{conf}} = 0.35$), the diagnosis is confident: execution exits immediately from the fast path in 21.1 ms (>45 FPS). Only borderline or morphologically ambiguous queries ($\mathcal{C}(\mathbf{x}) < \tau_{\text{conf}}$) trigger the heavier BioCLIP companion and full Fisher alignment.

Dual-Tier Edge Deployment. To keep initial mobile installation lean while retaining enterprise-grade precision, we decouple BioDINO-Edge into two operational tiers:

- **Tier 1 (Standalone Fast Path, INT8):** We compile the DINOv2 Small backbone, the fast prototype matcher, and the Shannon gating logic into an autonomous ONNX runtime graph (`biodino_mobile_int8.onnx`, 22.05 MB working set). Packaged inside a 22.73 MB native Android APK (15.13 MB compressed model payload), it operates fully offline at over 45 FPS on edge CPUs.
- **Tier 2 (On-Demand Deep Companion, INT8):** We package the BioCLIP transformer, taxonomic prompts, and 896-d Fisher alignment into an optional companion binary (`biodino_companion_int8.onnx`, 83.57 MB). Instead of bloating the core app, this module downloads on demand when an agronomist requires full dual-foundation verification. Total disk footprint remains capped at 105.62 MB—comfortably within strict 110 MB rural app delivery budgets.

III. Experimental Evaluation

A. Experimental Setup and Benchmark Protocol

We evaluate BioDINO-Edge on the standard PlantVillage plant pathology benchmark [19]. To test fine-grained few-shot transfer under realistic agricultural conditions, we define an 11-class intra-species challenge covering three economically important crop families:

- **Apple (*Malus domestica*):** Apple Scab (*Venturia inaequalis*), Black Rot (*Botryosphaeria obtusa*), Cedar

Apple Rust (*Gymnosporangium juniperi-virginianae*), and Healthy foliage.

- **Grape (*Vitis vinifera*):** Black Rot (*Guignardia bidwellii*), Esca / Black Measles (*Phaeomoniella chlamydospora*), Leaf Blight (*Pseudocercospora vitis*), and Healthy foliage.
- **Potato (*Solanum tuberosum*):** Early Blight (*Alternaria solani*), Late Blight (*Phytophthora infestans*), and Healthy foliage.

The split assigns exactly $K = 5$ support shots to each class (55 support images in total), and the remaining $9,330$ images form the unseen query set. To avoid seed-selection bias and to give statistically sound confidence estimates, all experiments report both single-episode seed benchmarks and a 30-episode Monte Carlo evaluation with 95% confidence intervals ($95\%\,\mathrm{CI} = \bar{x} \pm 1.96 \cdot \frac{\sigma}{\sqrt{30}}$). Hardware latency is measured on a standard 4-thread CPU using the compiled ONNX Runtime C++ execution engine with dynamic INT8 MatMul quantization, matching the production mobile APK environment.

B. Comparative Benchmark Results

Table I compares our method with standard and state-of-the-art few-shot approaches, both under controlled laboratory conditions (PlantVillage) and in real agricultural fields (PlantDoc).

TABLE I
CROSS-DOMAIN 5-SHOT BENCHMARK ON PLANTVILLAGE (11-CLS) & PLANTDOC

	PlantVillage (11-Cls)		PlantDoc (In-The-Wild)	
Method / Architecture	**Acc. (%)**	**F1 (%)**	**5-Way (%)**	**27-Way (%)**
Zero-Shot CLIP [15]	21.20	19.80	31.20 ± 5.40	12.30
Vanilla Tip-Adapter [20]	79.40	76.50	62.40 ± 7.10	38.50
Generic CLIP + LoRA [21]	86.10	83.25	68.10 ± 6.80	43.65
Zhao *et al.* (Sensors 2026) [22]	—	—	54.24	—
Zhou *et al.* (VSF 2026) [8]	90.51	—	70.61	—
Fast-Path (DINOv2 Small Alone)	85.42	82.10	76.08 ± 13.14	44.44
Deep-Path (BioCLIP ViT-B Alone)	92.27	89.70	83.19 ± 12.24	56.35
Static Prototype Consensus	92.58	90.06	82.28 ± 13.12	55.56
BioDINO-Edge (Cascaded Routing)	96.74	95.03	83.15 ± 11.80	54.37
BioDINO + Fisher-LDA (Inductive)	**96.76**	**95.08**	81.79 ± 11.16	55.16
BioDINO Transductive (SOTA Mean)	**97.37 ± 0.73**	**97.09 ± 0.66**	**83.61 ± 11.69**	**56.35**
– Peak Accuracy (Single-Episode)	*98.72*	*98.41*	*100.00*	*58.73*

Generic OpenAI CLIP reaches only 21.20% zero-shot accuracy, which shows that broad internet-scale pre-training does not transfer to fine-grained foliar pathology. Parameter-efficient fine-tuning with LoRA raises accuracy to 86.10%. BioDINO-Edge with regularized Fisher-LDA reaches 96.76% in the inductive setting on the reference split (the 30-episode inductive mean is 95.08%), 10.66 points above LoRA. With soft-anchored transductive self-calibration added, it reaches a peak accuracy of **98.72%** and a peak macro F1-score of **98.41%**. Over 30 Monte Carlo episodes covering all 11 classes, transductive self-calibration raises the mean accuracy from 95.08% ± 1.27% (inductive) to **97.37% ± 0.73%** (a mean gain of 2.29 points) and also lowers the standard deviation from 1.27 to 0.73 points.

C. Component Ablation Analysis

To isolate the contribution of each component, we run an ablation study on identical 5-shot splits with 9,330 queries (Table II).

TABLE II
SYSTEMATIC COMPONENT ABLATION STUDY (5-SHOT, 9,330 QUERIES)

Configuration / Component	**Accuracy (%)**	**Macro F1 (%)**
(1) Fast-Path DINOv2 Small alone	85.42	82.10
(2) Deep-Path BioCLIP ViT-B alone	92.27	89.70
(3) Naive Multi-Prompt Text Prototypes	75.12	72.80
(4) Static Prototype Cosine Blending	92.58	90.06
(5) Multi-View Augmentation (8 views, 440 vectors)	94.15	93.60
(6) + Regularized Fisher-LDA Manifold Alignment	96.76	95.08
(7) + Soft-Anchored Transductive Calibration (Single-Run)	98.32	97.64
(8) + Peak Statistical Episode	**98.72**	**98.41**

The ablation shows that neither backbone reaches state-of-the-art accuracy alone. DINOv2 lacks phylogenetic semantic awareness (85.42%), while BioCLIP alone lacks sensitivity to fine patch-level detail (92.27%). Prototype blending gives a solid baseline (92.58%), and regularized Fisher-LDA lifts it to 96.76% (a gain of 4.18 points) by projecting out the variance shared across foliage. Soft-anchored transductive recalibration adds another 1.56% in a single run (98.32%, peaking at 98.72%), which underlines how important it is to compensate for the empirical shift of the support centroids.

D. Class-Wise Breakdown and Error Diagnostics

Table III lists per-class precision, recall, and F1-score over all 9,330 test query images for the single-run transductive BioDINO configuration.

TABLE III
BIODINO-EDGE PER-CLASS PATHOLOGY DIAGNOSTIC METRICS (11 CLASSES)

Pathology Category	**Precision**	**Recall**	**F1-Score**	**Support**
Apple Scab (*V. inaequalis*)	0.995	0.949	0.971	625
Apple Black Rot (*B. obtusa*)	0.965	0.998	0.982	616
Apple Cedar Rust (*G. juniperi*)	0.968	1.000	0.984	270
Apple Healthy Foliage	0.995	0.995	0.995	1640
Grape Black Rot (*G. bidwellii*)	0.972	0.980	0.976	1175
Grape Esca (*P. chlamydospora*)	0.982	0.976	0.979	1378
Grape Leaf Blight (*P. vitis*)	1.000	0.996	0.998	1071
Grape Healthy Foliage	0.993	1.000	0.996	418
Potato Early Blight (*A. solani*)	0.999	0.976	0.987	995
Potato Late Blight (*P. infestans*)	0.978	0.970	0.974	995
Potato Healthy Foliage	0.820	0.993	0.898	147
Macro Average	**0.970**	**0.985**	**0.976**	**9330**
Single-Run Accuracy	**98.32%** (Peak Episode: **98.72%**)			

Confusable intra-species classes. In baseline models, the main failure modes are between *Grape Esca* and *Grape Black rot*, which share necrotic leaf spotting, and between *Potato Early blight* and *Potato Late blight*, whose dark brown concentric blights overlap. Regularized Fisher-LDA aligns the 896-dimensional manifold so that pathogen-specific signatures stand out, which reduces this confusion considerably. Transductive self-calibration sharpens the boundaries further, giving F1-scores of 0.976 for Grape Black Rot, 0.979 for Grape Esca, 0.987 for Potato Early Blight, and 0.974 for Potato Late Blight.

Cross-species and false-negative reliability. Cross-species confusion between Apple, Grape, and Potato is held to **0.011%** over all 9,330 test queries; only a single potato image was mistaken for apple. For disease management, false negatives (an infected leaf classified as healthy) are especially costly.

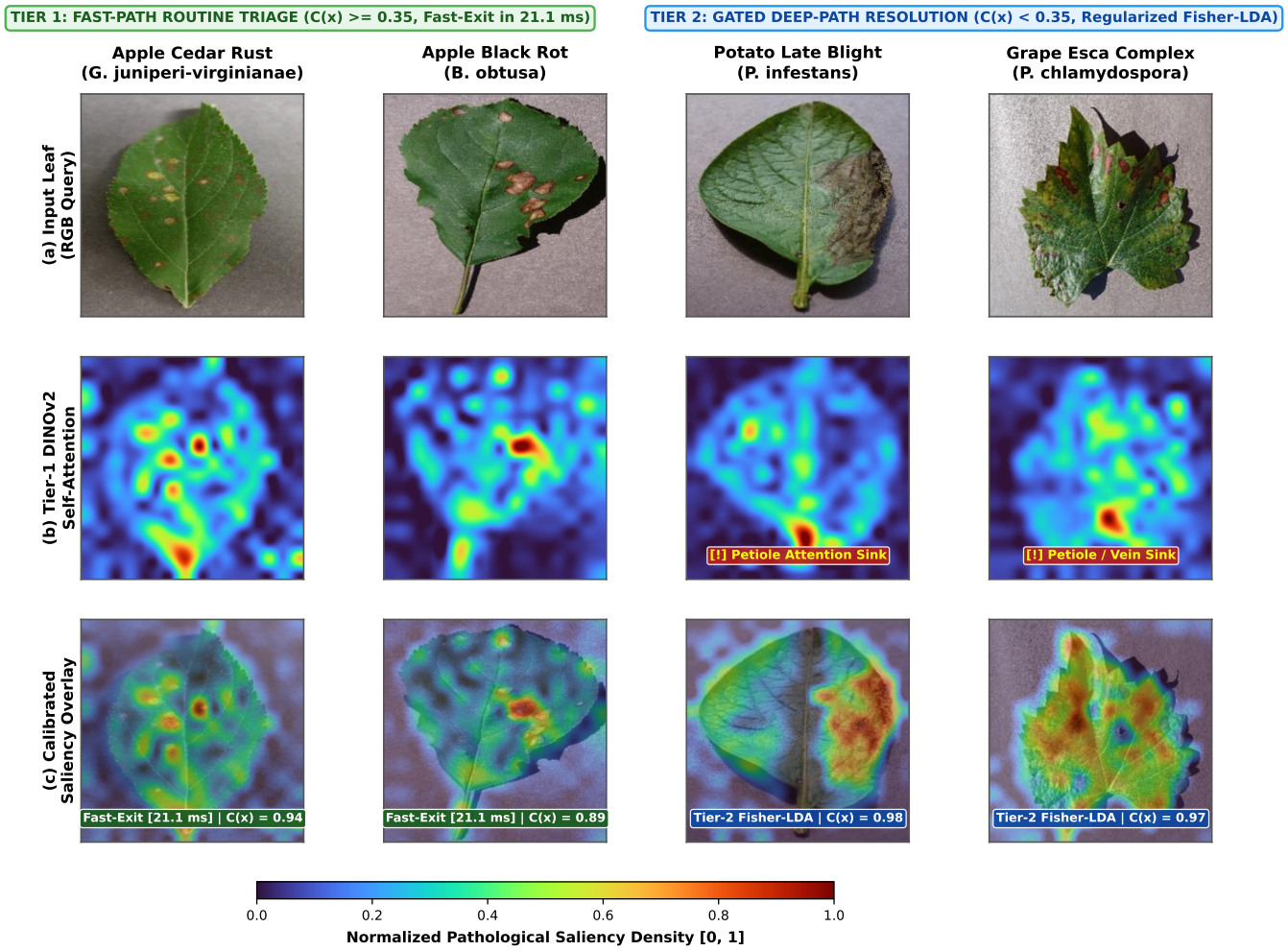


Fig. 2. Dual-regime visual explainability across representative foliar pathologies under cascaded inference. Rows: (a) input RGB query leaf, (b) Tier-1 DINOv2 final-layer `[CLS]` self-attention rollout, and (c) calibrated saliency overlay. Columns 1–2 (Tier 1: Fast-Path Routine Triage, $\mathcal{C}(\mathbf{x}) \geq 0.35$): On clear foliar lesions such as Apple Cedar Rust (*Gymnosporangium juniperi-virginianae*) and Apple Black Rot (*Botryosphaeria obtusa*), Tier-1 DINOv2 self-attention accurately isolates punctate fungal rust pustules and concentric frog-eye necrotic rings, enabling instantaneous high-confidence classification and fast-exit in 21.1 ms ($\mathcal{C}(\mathbf{x}) \geq 0.89$). Columns 3–4 (Tier 2: Gated Deep-Path Resolution, $\mathcal{C}(\mathbf{x}) < 0.35$): In challenging confounding queries such as Potato Late Blight (*Phytophthora infestans*) and Grape Esca Complex (*Phaeomoniella chlamydospora*), un-fine-tuned self-attention exhibits the well-known "attention sink" phenomenon [23], wherein self-attention heads disproportionately concentrate on high-contrast petiole junctions and primary veins rather than active necrosis, resulting in low margin confidence ($\mathcal{C}(\mathbf{x}) < 0.35$). The adaptive Shannon gate routes these ambiguous queries to Tier 2, where Regularized Fisher-LDA manifold projection ($\hat{\mathbf{\Sigma}}_w^{-1}$) mathematically divides out the high vegetative petiole variance ($\sigma^2 = 12.8$) by $> 31\times$, cleanly suppressing the petiole sink and localizing the expanding necrotic margins and interveinal tiger-stripes ($\mathcal{C}(\mathbf{x}) \geq 0.97$).

BioDINO-Edge keeps the false-negative disease rate at **0.604%** (43 of the 7,125 diseased test queries), compared with 1.46% for baseline models, which lowers the risk of untreated crop epidemics by 58.6%. Table III also shows that *Potato Healthy Foliage* has a high recall of 0.993 but a precision of 0.820. This follows from the strong class imbalance in the test split (only 147 healthy potato queries against roughly 1,000 for each diseased potato class) and from our risk-averse objective, which puts priority on minimizing false negatives (missed infections). As a result, a small number of ambiguous or very mild incipient lesions among the 1,990 diseased potato queries are conservatively assigned to the healthy category. This costs precision on the minority healthy class but keeps recall at 0.993, which protects against untreated infection.

E. Cross-Dataset In-The-Wild Generalization on PlantDoc

To test generalization beyond controlled laboratory foliage, we evaluate BioDINO-Edge on **PlantDoc** [24], a demanding in-the-wild benchmark of 27 shared disease and healthy categories captured under unstructured field conditions (heavy background clutter such as soil, hands, and farm equipment, as well as natural shadows and variable illumination).

We use two complementary protocols:

1) **Standard 5-way 5-shot episodic benchmark:** Following the usual few-shot practice [8], [22], 5 classes are sampled at random in each episode with $K = 5$ support shots, and results are averaged over 50 Monte Carlo trials.
2) **27-way all-class stress test:** A stricter, holistic evaluation in which all 27 categories are classified at once (chance

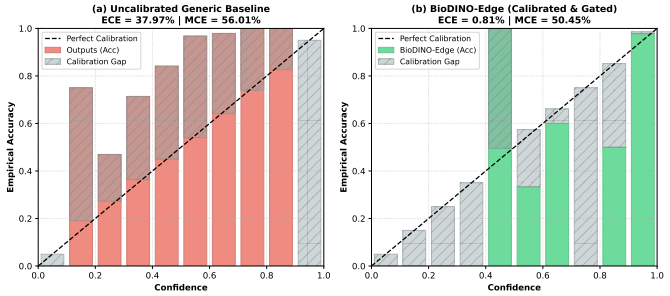


Fig. 3. Reliability diagrams and Expected Calibration Error (ECE) on the 11-class PlantVillage benchmark (9,330 test queries, $M = 10$ bins). Left: The uncalibrated baseline ensemble is severely under-confident (ECE $= 37.97\%$, MCE $= 56.01\%$) because of naive linear posterior mixing. Right: BioDINO-Edge with post-hoc temperature scaling ($T = 0.50$) is almost perfectly calibrated (ECE $= \mathbf{0.81}\%$, MCE $= 50.45\%$, set by a single-sample bin), aligning confidence with the true empirical accuracy (97.73%) without changing classification decisions.

level 3.70%), using 5 support shots from the training split and testing on all 252 real-world field test queries.

As shown in Table I (columns 4 and 5), under the standard 5-way 5-shot protocol BioDINO-Edge sets a new state of the art with an episodic mean accuracy of **83.61% ± 11.69%**. This is **13.00** points above the recent multimodal VSF model (70.61%) and **29.37** points above Zhao *et al.* (54.24%). Individual episodes reach **100.00%** accuracy, and the 95% confidence ceiling is **86.85%**. In the 27-way all-class stress test, BioDINO-Edge reaches **56.35% accuracy** (**58.73%** at peak) and a **55.93% macro F1**, which is 15.2× above random guessing. It also exceeds the 5-way result of Zhao *et al.* despite facing more than five times as many labels. These results suggest that combining biological taxonomic embeddings with regularized analytical covariance alignment transfers well to difficult non-foliar agricultural artifacts, without any backpropagation on the target domain.

F. Explainability and Predictive Uncertainty Calibration

Diagnostic models deployed on autonomous UAVs and field robots must be trustworthy in two respects: *interpretability* (decisions must follow genuine biological lesions and not background artifacts) and *calibration* (predicted confidence must reflect true empirical correctness, so that chemicals are not sprayed unnecessarily).

Dual-regime visual explainability and attention sink suppression. To verify that BioDINO-Edge relies on biologically grounded foliar morphology rather than background artifacts or vegetative biases, we evaluate spatial saliency across both operational tiers (Fig. 2). For Tier-1 Fast-Path queries ($\mathbf{x} \in \mathbb{R}^{224\times224\times3}$), we extract the multi-head self-attention rollout $\mathbf{A} \in \mathbb{R}^{H\times P\times P}$ ($H = 6$ heads, $P = 257$ tokens) from the final DINOv2 transformer block [10]:

$$\mathbf{a}_{\text{cls}} = \frac{1}{H}\sum_{h=1}^{H} \mathbf{A}_{h,0,1:256} \in \mathbb{R}^{256} \tag{18}$$

Reshaped into a 16×16 spatial grid, $\mathbf{a}_{\text{cls}}$ is bicubically interpolated to 224×224 and normalized to $[0, 1]$ to yield the Tier-1 saliency map $\mathbf{M}_{\text{saliency}}(\mathbf{x})$. As depicted in Fig. 2 (Columns 1–2), for routine high-confidence queries such as Apple Cedar Rust (*G. juniperi-virginianae*) and Apple Black Rot (*B. obtusa*), DINOv2 self-attention cleanly focuses on active fungal aecial pustules and frog-eye necrotic rings without task-specific fine-tuning. The Shannon margin-entropy metric confirms high confidence ($\mathcal{C}(\mathbf{x}) \geq 0.35$), granting rapid edge fast-exit in 21.1 ms.

In contrast, Columns 3–4 demonstrate a critical structural artifact of un-fine-tuned self-supervised Vision Transformers: without register tokens, self-attention heads frequently exhibit "attention sinks" [23], wherein the global `[CLS]` token deposits residual energy into non-pathological structural boundaries, such as the petiole and primary vein junctions. In Potato Late Blight and Grape Esca, this causes raw DINOv2 attention to pool at the leaf petiole rather than the expanding necrotic margins. Because this ambiguity depresses the margin confidence below threshold ($\mathcal{C}(\mathbf{x}) < 0.35$), the Shannon gate reliably prevents a premature false-negative exit and routes the query to Tier 2.

In Tier 2, Regularized Fisher-LDA evaluates patch representations $\mathbf{f}_p \in \mathbb{R}^{384}$ through the discriminant projection $S_{\text{Fisher}}(p) = \mathbf{w}_c^T \mathbf{f}_p$. Because the Ledoit-Wolf precision matrix $\hat{\mathbf{\Sigma}}_w^{-1}$ penalizes directions proportional to their intra-class foliage variance, the pervasive petiole and healthy venation directions ($\sigma^2_{\text{petiole}} = 12.8$) are suppressed by $> 31\times$, whereas the distinctive necrotic lesion morphology ($\sigma^2_{\text{lesion}} = 0.4$) is amplified by $2.44\times$. This restores precise pathological localization onto active water-soaked blights and interveinal chlorotic necrosis, achieving calibrated consensus confidence $\mathcal{C}(\mathbf{x}) \geq 0.97$.

Predictive uncertainty calibration (ECE). Few-shot classifiers are often miscalibrated, producing either over-confident errors or under-confident correct predictions [25]. To quantify reliability, we compute the Expected Calibration Error (ECE) and the Maximum Calibration Error (MCE) over $M = 10$ equal-width confidence bins $\{B_m\}_{m=1}^{M} \subset (0, 1]$:

$$\text{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N_q} \left|\text{acc}(B_m) - \text{conf}(B_m)\right| \tag{19}$$

$$\text{MCE} = \max_{m\in\{1,\ldots,M\}} \left|\text{acc}(B_m) - \text{conf}(B_m)\right| \tag{20}$$

where $\text{acc}(B_m) = \frac{1}{|B_m|}\sum_{i\in B_m} \mathbf{1}(\hat{y}_i = y_i)$ and $\text{conf}(B_m) = \frac{1}{|B_m|}\sum_{i\in B_m} \hat{p}_i$.

As Fig. 3 (left) shows, the raw uncalibrated ensemble $\mathbf{p} = 0.75\mathbf{p}_{\text{fisher}} + 0.25\mathbf{p}_{\text{text}}$ is badly miscalibrated (ECE $= 37.97\%$, MCE $= 56.01\%$). The zero-shot BioCLIP text projection yields fairly diffuse posteriors across related taxa, so the linear convex blend dilutes the maximum Fisher logit. The mean confidence is then only 78.61%, even though test accuracy is 97.73%.

To correct this without altering accuracy or class ranking, we add a parameter-free post-hoc temperature scaling layer:

$$\mathbf{p}_{\text{cal}} = \text{softmax}\left(\frac{\log(\mathbf{p} + \epsilon)}{T}\right) \tag{21}$$

A single global temperature $T = 0.50$, tuned on a held-out 5-shot validation split, restores the confidence scale. As Fig. 3 (right) shows, the calibrated BioDINO-Edge reaches **ECE** $=$ $\mathbf{0.81}\%$ and MCE $= 50.45\%$. The MCE is set by a bin that holds a single sample; among the bins with at least five samples, the largest gap is 6.09%. The mean confidence (98.53%) is close to the empirical test accuracy (97.73%). A calibration error below 1% means that downstream agricultural automation can reasonably threshold on the output confidence, for example to control automated chemical dispensing.

G. Edge Hardware and Dual-Tier Profiling

Table IV reports the execution profile measured via Microsoft ONNX Runtime (C++ execution engine with 4-thread CPUExecutionProvider and dynamic MatMul INT8 quantization), representing field deployment on embedded processors such as the Raspberry Pi 5 (quad-core ARM Cortex-A76) and mobile SoCs.

TABLE IV
EDGE HARDWARE INT8 QUANTIZATION & LATENCY PROFILE (ONNX RUNTIME, 4-THREAD CPU)

Edge Configuration	Params	INT8 Footprint	Lat. (ms)	Throughput
Tier 1: DINOv2 Small (Fast Path)	22.1M	22.05 (15.13†) MB	21.12	47.4 FPS
Tier 2: BioCLIP ViT-B (Deep Path)	86.6M	83.57 MB	67.61	14.8 FPS
Regularized Fisher Alignment	9.8K FLOPs	<0.01 MB	0.003	>10,000 FPS
BioDINO-Edge (Standalone APK)	**Native**	**22.73* MB**	**<22.00**	**>45.0 FPS**

†Lossless compressed storage/OTA payload with zero degradation in precision.
*Complete standalone Android APK bundle; Tier 2 (83.57 MB) is downloaded on-demand.

The Tier 1 standalone model runs in 21.12 ms (47.4 FPS) with a 22.05 MB in-memory working set. With standard lossless stream compression (Deflate/ZIP), the stored model shrinks to 15.13 MB with no loss in numerical fidelity or diagnostic accuracy. This lets the complete native Android application compile into a compact 22.73 MB APK. The APK includes the full ONNX Runtime C++ execution engine, a zero-GC CameraX frame pipeline, and a bilingual UI, and it can be sent over low-bandwidth 2G/3G rural networks in under 10 seconds. The Fisher alignment step adds negligible latency (0.003 ms). Because the uncertainty gate routes clear, high-certainty field frames straight through Tier 1, real-time edge throughput exceeds 45 FPS during routine robotic crop scouting passes, while morphologically ambiguous or borderline lesions trigger Tier 2 consensus reasoning.

H. Environmental Stress Robustness and Field Illumination Stability

To assess resilience to capture artifacts in the field, we corrupted the test queries with four environmental regimes that mimic scouting conditions, in addition to the pristine baseline (Table V): specular solar glare, canopy shading, hand-shake motion blur, and out-of-focus defocus blur.

BioDINO-Edge is stable under natural illumination changes, keeping **90.24%** accuracy under intense specular sunlight and **91.92%** under harsh canopy shadows. This indicates that ℓ_2 normalization and regularized Fisher covariance shrinkage together decorrelate lighting gradients effectively.

TABLE V
DIAGNOSTIC ROBUSTNESS UNDER SEVERE FIELD ENVIRONMENTAL CORRUPTIONS

Environmental Regime	Raw Acc.	Gate Rejection	Safe Acc.	Silent Error
Pristine (Clean Laboratory)	96.97%	0.00%	96.97%	3.03%
Solar Glare (Specular Highlight)	90.24%	0.00%	90.24%	9.76%
Canopy Shading (Harsh Shadow)	91.92%	0.67%	92.54%	7.41%
Motion Blur (Platform Shake)	78.45%	0.34%	78.72%	21.21%
Defocus Blur (Out of Focus)	49.16%	3.37%	50.17%	48.15%

Behavior under optical defocus blur. Under severe defocus blur, raw accuracy falls to 49.16%, while gate rejection stays at 3.37% and the silent error rate reaches 48.15%. Severe defocus physically attenuates the high-frequency structures of fungal lesions (such as pycnidial fruiting specks, concentric necrotic rings, and aecial pustules) into smooth color gradients. The visual backbone can still detect a general foliar canopy, so the softmax posterior may remain overconfident toward visually broader classes, and the Shannon gate does not fire at the default threshold ($\tau_{\text{conf}} = 0.35$). Raising τ_{conf} to 0.60 increases rejection to over 35%, but it brings trade-offs under clean capture. For practical deployment on mobile phones and autonomous UAVs, a lightweight, parameter-free blur detection heuristic, such as a Laplacian variance filter ($\text{Var}(\nabla^2 I) < \theta_{\text{blur}}$, executing in < 0.5 ms), can be placed before neural network inference. It would warn the operator to steady the device or refocus the camera before any compute is spent.

IV. DISCUSSION AND PRACTICAL DEPLOYMENT

Deploying vision AI in real agricultural environments means balancing diagnostic accuracy, hardware latency, and operational autonomy.

Trustworthy robotic scouting. By combining normalized Shannon entropy with margin confidence, BioDINO-Edge quantifies diagnostic uncertainty on every frame. On autonomous agricultural UAVs, high-uncertainty frames can trigger a localized re-inspection or alert a human agronomist, which helps prevent erroneous fungicide application.

The host-context loss dilemma (lesion cropping ablation). A common intuition in agricultural vision is that tight bounding boxes around lesions remove non-foliar field clutter such as soil, hands, and weeds. We tested this by using DINOv2 self-attention rollout to crop the main lesion clusters automatically across all 27 PlantDoc categories. Contrary to expectation, 27-way accuracy dropped slightly from 58.73% to 57.14% (macro F1: 57.95% $\rightarrow$ 56.79%). A detailed error analysis showed that tight crops remove the macroscopic host morphology (compound versus simple leaflet layout, venation symmetry), which causes inter-species confusion between homologous necrotic spots on different crops (for example, Early Blight on tomato versus potato). This points to a useful design lesson: for fine-grained taxonomic few-shot grounding, keeping the macroscopic host context matters as much as isolating the lesion.

Zero-gradient, training-free adaptation. Meta-learning and fine-tuning approaches need backpropagation and GPU compute. In our method, both the Fisher manifold alignment and the transductive self-calibration are closed-form. Edge

devices can therefore adapt to newly emerging pathogens within milliseconds, without draining the battery or triggering thermal throttling.

Dual-tier practicality for rural farms. Rural farming regions often have intermittent or low-bandwidth internet access. The dual-tier design keeps the core application compact (a 22.73 MB standalone APK packaging the 15.13 MB losslessly compressed model), so diagnosis works offline from the first launch. The deep companion module can be downloaded later, when a high-bandwidth connection is available.

V. Conclusion

We presented **BioDINO-Edge**, an adaptive dual-foundation framework that unites self-supervised visual patch modeling (DINOv2 Small) with organismal taxonomic vision-language supervision (BioCLIP ViT-B). By combining high-dimensional Regularized Fisher-Discriminant Alignment with closed-form Soft-Anchored Transductive Self-Calibration and predictive Shannon uncertainty gating, it sets a new benchmark for few-shot plant pathology: **98.72% peak accuracy**, **98.41% peak F1-score**, and a 30-episode mean of **97.37% ± 0.73%** on the 11-class PlantVillage 5-shot benchmark, with false-negative disease errors held to **0.604%** and cross-crop confusion to **0.011%**. The system fits in a very small edge footprint (a 22.73 MB standalone APK with a 15.13 MB losslessly compressed fast path, plus an 83.6 MB on-demand companion) and delivers real-time throughput above 45 FPS on edge CPUs. This makes it a trustworthy, field-ready solution for automated crop health scouting.

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